

House Price Prediction Using Machine Learning: A Study on the Impact of Spatial and Environmental Features

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Abstract—This study investigates the problem of predicting house prices using machine learning techniques. The dataset used in this study contains various housing features, including structural characteristics, environmental factors, and spatial information such as city and waterfront. A complete data modelling pipeline was applied, including data preprocessing, feature engineering, and model training. Several regression models, including Linear Regression, Random Forest, and Gradient Boosting, were implemented and compared. The results show that ensemble models outperform linear models due to their ability to capture non-linear relationships. In addition, spatial and environmental features such as city and waterfront were found to have a significant impact on house prices. These findings highlight the importance of incorporating both structural and spatial features in house price prediction.

Index Terms—house price prediction, machine learning, regression, spatial features, environmental factors

I. INTRODUCTION

House price prediction is a key task in the real estate industry. Real estate plays an important role in the global economy and affects many stakeholders, including buyers, sellers, investors and policymakers [1]. Moreover, in reality, house prices are influenced by a wide range of factors, including complex geographical factors, local regulatory policies and neighborhood characteristics. Hence, accurate property valuation becomes particularly challenging yet essential for supporting informed decision-making, ensuring economic transparency and maintaining economic stability. Specifically, as a highly differentiated product, the value of real estate is shaped by both internal factors (such as building quality and layout) and external environment attributes (such as geographical location and employment accessibility) [2]. Traditional approaches, which highly rely on expert judgement or simplified statistical methods, such as averaging out decades of data or the neighborhood data, often struggle to capture the intricate interactions among these factors. Combined with limited market information and low transaction frequency, such methods are prone to subjective bias and inconsistent valuations [1].

To address these limitations, machine learning has been increasingly adopted for house price prediction. Unlike traditional statistical approaches, these methods are capable of capturing complex and non-linear relationships between house features, thereby achieving significantly higher predictive accuracy. A wide range of algorithms have been explored in this domain, including linear regression, decision trees, random forest, and gradient boosting models. However, despite these advances, no single model can consistently achieve the best performance across all scenarios. Each model type has inherent trade-offs: linear models offer interpretability but struggle with nonlinear patterns [3]; tree-based models capture complex interactions but risk overfitting [4], [5]; and neural networks provide flexibility at the cost of requiring careful and delicate architecture design and hyperparameter tuning. To overcome these individual limitations, ensemble learning methods have been introduced into house price prediction. Among them, stacking is particularly promising, as it uses a meta-learner to integrate predictions from heterogeneous base models through a multilayer architecture. By combining diverse models that capture different aspects of the data, stacking can correct individual prediction biases and improve overall accuracy.

In order to overcome the limitations of a single prediction model, the integrated learning method has been widely introduced into the field of house price prediction. Among them, Stacking uses a meta-learner to integrate predictive output from heterogeneous learners (such as decision trees, random forests, gradient boosting, and neural networks) by building a multi-layer learning architecture. This mechanism can capture and correct the prediction bias of different algorithms and significantly improve the accuracy and robustness of the general prediction through the synergistic complementary effect [6], [7].

Despite its potential, the systematic application of stacking ensembles to house prediction remains limited [8]. Most existing studies focus on individual models or simpler ensemble techniques such as Bagging and Boosting [5], with relatively few investigating stacking in this context. Moreover, the studies that do apply stacking have reported mixed findings:

while some improvements in accuracy have been observed, the gains are often marginal and come at a high computational cost, raising questions about the methods' general applicability in the real market [9]. More critically, existing stacking-based studies tend to treat the ensemble as a black box, with insufficient analysis of how each base learner contributes to the overall predictions. This limits both the theoretical understanding and practical adoption of the methods in housing valuation [9]. These gaps motivate the present study, which aims to evaluate whether a stacking ensemble can consistently outperform individual models and to analyse the specific role of each base model within the ensemble.

To investigate these research gaps, this study used the King County housing dataset obtained from Kaggle, which contains 4600 house transactions recorded between May 2014 and May 2015. The dataset included 18 features describing each property, covering structural attributes (e.g., bedrooms, bathrooms, and living area), as well as spatial and environmental factors (e.g., waterfront and view). The target variable is the house sale price. Hence, this dataset is particularly suited for evaluating stacking ensembles as various feature types allow different base models to leverage different aspects of the data.

Specifically, this study addresses the following research questions:

- How do individual machine learning models compare in predicting house prices?
- Can a stack ensemble that combines the base models outperform the best-performing individual models?
- Which base models contribute most to the stacking ensemble's performance?

II. LITERATURE REVIEW

With the rapid development of data-driven technology, machine learning methods are increasingly used in house price prediction. These methods can capture the complex and nonlinear relationship between house features, thus significantly improving the predictive performance [10]. Researchers have verified a variety of algorithms, including traditional linear regression, decision tree, random forest and gradient boosting model [1], [4], [11]. Compared with traditional parameter models (such as hedonic regression), these machine learning algorithms show stronger robustness and prediction accuracy when processing large-scale and high-dimensional real estate data, especially in capturing complex nonlinear relationships.

However, although the machine learning model shows good performance, no model can always maintain the best performance in all scenarios. Different models have their own advantages and disadvantages. Linear models (such as traditional hedonic regression) are simple, computationally efficient and easy to explain, but it is difficult to capture complex nonlinear relationships in housing data [3]. Tree-based models (such as decision trees and random forests) can capture nonlinear interactions between features and process high-dimensional data [4], [5]. However, while improving the prediction accuracy, such models also face the challenge of easy overfitting and lower model interpretability than traditional linear models [5].

Neural network models can learn highly complex nonlinear patterns and interactions, but their performance is highly dependent on network structure design, and usually requires fine parameter tuning through trial and error [3]. Support vector regression (SVR) can effectively process nonlinear relationships in house price data in higher-dimensional feature space by introducing kernel functions [12]. However, this method has significant limitations in practical application: on the one hand, its predictive performance shows high sensitivity to the selection of penalty parameters C , ε -insensitive zone and nuclear function parameters [12]. On the other hand, as its computational complexity increases significantly with the increase of the sample size, its calculation cost is often higher than that of tree-based models (such as random forests) when processing large-scale real estate data sets. While linear models provide interpretability, and tree-based models capture nonlinear interactions, their performance trade-offs highlight the absence of a universally optimal model for house price prediction.

Overall, these studies demonstrate that while machine learning methods outperform traditional approaches, their performance varies significantly depending on the model choice and data characteristics.

To further improve predictive performance, ensemble learning methods have been widely adopted [13]. Techniques such as bagging and boosting aim to improve model stability and reduce variance by combining multiple learners [14]. For example, random forests and gradient boosting models have demonstrated strong performance in house price prediction tasks due to their ability to aggregate multiple decision trees and capture complex feature interactions [11]. Furthermore, stacking combines heterogeneous base models using a meta-learner, allowing the integration of complementary strengths from different algorithms. This multi-layer learning framework has the potential to improve both predictive accuracy and robustness by correcting individual model biases [15].

However, despite these advances, existing studies often lack systematic and controlled comparisons between different models under consistent experimental settings, making it difficult to draw generalizable conclusions about their relative performance [8]. Moreover, the application of stacking ensembles in house price prediction remains relatively underexplored [16], and there is limited analysis of how individual base learners contribute to the overall performance of the ensemble.

Therefore, there remains a research gap in systematically evaluating whether stacking ensembles can consistently outperform individual models, as well as in understanding the role of each base learner within the ensemble. This study aims to address this gap by providing a comprehensive comparison of multiple machine learning models and analysing the effectiveness of stacking ensembles in real-world house price prediction tasks.

III. METHODOLOGY

A. Overall Modelling Pipeline

The overall modelling pipeline consists of the following stages:

- 1) Data cleaning and preprocessing (outlier removal, encoding, scaling)
- 2) Feature engineering (temporal, structural, and renovation-related features)
- 3) Base model training (Linear Regression, Random Forest, XGBoost, MLP)
- 4) Generation of Out-of-Fold (OOF) predictions using 5-fold cross-validation
- 5) Construction of meta-features for stacking
- 6) Training of a meta-learner (Ridge Regression)
- 7) Model evaluation using RMSE, R^2 , and cross-validation

B. Dataset Description

The dataset used in this study is the King County housing dataset obtained from Kaggle (shree1992/housedata). It contains approximately 4,600 residential property records with 18 features, including structural attributes (e.g., bedrooms, bathrooms, and living area) as well as spatial and environmental variables (e.g., waterfront and view). The target variable is the house sale price.

This dataset is suitable for house price prediction as it captures both physical characteristics and location-related factors, enabling the modelling of complex relationships in real estate pricing.

C. Data Preprocessing

Data preprocessing was conducted to improve data quality, reduce noise, and ensure compatibility with machine learning models.

First, the dataset was examined for missing values and data quality issues. No significant missing values or invalid entries were identified; therefore, no imputation or removal based on missing data was required.

Second, outliers in the target variable (price) were addressed using the Interquartile Range (IQR) method. Observations lying outside the range defined by $Q1 - 1.5 \times IQR$ and $Q3 + 1.5 \times IQR$ were removed. This step reduces the influence of extreme values that could disproportionately affect regression models and evaluation metrics such as RMSE and R^2 .

Third, irrelevant and high-cardinality features, including *street*, *statezip*, *country*, and the raw *date* field, were removed after feature extraction. These variables contain a large number of unique values and may introduce noise or lead to overfitting without contributing meaningful generalisable patterns.

Fourth, categorical variables were transformed into numerical representations using one-hot encoding. In particular, features such as *city* and *season* were converted into binary indicator variables. Additionally, boolean features were converted into integer format (0/1) to ensure numerical consistency across the dataset.

Fifth, a structured modelling dataset (*df_model*) was constructed by combining the cleaned and engineered features, ensuring that only numerical variables were retained for model training.

Finally, feature scaling was applied using *StandardScaler*. All numerical features were standardised to zero mean and unit variance before model training. This step is particularly important for models sensitive to feature scale, such as linear regression and multi-layer perceptron (MLP), as it ensures stable optimisation and prevents features with larger magnitudes from dominating the learning process.

D. Feature Engineering

Feature engineering was performed to enhance the representational capacity of the dataset and to better capture meaningful relationships relevant to house price prediction.

First, temporal information was extracted from the transaction date. The original *date* feature was converted into datetime format, from which *year* and *month* features were derived. These variables enable the models to capture temporal trends and potential seasonal patterns in housing prices.

Second, a new feature *house_age* was constructed as the difference between the transaction year and the construction year (*yr_built*). This transformation provides a more meaningful representation of property depreciation compared to using raw construction year values.

Third, renovation-related features were introduced to capture both the existence and recency of property improvements. A binary feature *is_renovated* was created to indicate whether a property had undergone renovation. Additionally, the feature *year_since_renovation* was defined with conditional logic: properties without renovation were assigned a value of zero, while renovated properties were assigned the difference between the transaction year and the renovation year. This ensures consistency while preserving interpretability.

Fourth, seasonal effects were incorporated by mapping the *month* feature into four categories: spring, summer, autumn, and winter. These categorical features were later encoded into numerical variables during preprocessing, enabling the model to capture cyclical variations in housing demand.

Fifth, a composite feature *total_sqft* was constructed by summing *sqft_living* and *sqft_lot*. This feature provides a more holistic measure of property size. Empirical analysis using correlation metrics showed that *total_sqft* captures additional variance in house prices compared to using *sqft_living* alone, justifying its inclusion.

Finally, a logarithmic transformation of the target variable (*log_price*) was explored to reduce skewness and stabilise variance. Although this transformation improved the distributional properties of the target variable, the final models were trained using the original price variable to maintain interpretability of predictions in real-world units.

Overall, the feature engineering process was guided by both domain knowledge and empirical analysis, ensuring that the constructed features are meaningful, interpretable, and beneficial for model performance.

E. Model Design

To overcome the limitations of individual models and improve predictive performance, a two-level stacking ensemble architecture was designed. The overall framework integrates multiple heterogeneous base learners and a meta-learner, with additional enhancements including hyperparameter tuning, feature passthrough, and ablation analysis.

1) *Level 1: Base Learners*: Four heterogeneous machine learning models were selected as base learners:

- **Linear Regression (LR)**: Captures global linear relationships and serves as a simple and interpretable baseline.
- **Random Forest (RF)**: A bagging-based ensemble model that captures nonlinear feature interactions and is robust to noise and outliers.
- **Extreme Gradient Boosting (XGBoost)**: A boosting-based model that iteratively improves prediction accuracy by minimising residual errors and is well-suited for tabular data.
- **Multi-Layer Perceptron (MLP)**: A neural network model capable of learning complex nonlinear mappings between input features and house prices.

To improve model performance, hyperparameter tuning was applied, particularly for XGBoost using grid search with cross-validation. Key parameters such as learning rate, tree depth, and number of estimators were optimised. Additionally, the MLP model was stabilised using early stopping and controlled network architecture to improve convergence.

Each base model was first trained and evaluated individually on the test set to establish baseline performance, allowing a fair comparison with the stacking ensemble.

2) *Out-of-Fold Prediction Strategy*: A 5-fold cross-validation strategy was employed to generate Out-of-Fold (OOF) predictions for each base learner.

The training data was split into five folds. For each fold, a fresh clone of the model was trained on four folds and used to generate predictions on the remaining unseen fold. This ensures that each prediction is made on data not used during training, preventing data leakage.

The OOF predictions from all base models were then combined to form a new feature space for the meta-learner. In addition, test set predictions were generated by averaging predictions across folds.

3) *Level 2: Meta-Learner*: At the second level, a Ridge Regression model was used as the meta-learner. Compared to standard linear regression, Ridge regression introduces regularisation, which helps prevent overfitting when combining multiple model predictions.

Two stacking variants were implemented:

- **Standard Stacking**: using only base model predictions as input features.
- **Enhanced Stacking (Passthrough)**: combining base model predictions with the original input features.

The passthrough strategy allows the meta-learner to utilise both learned predictions and original feature information,

increasing model flexibility and improving predictive performance.

4) *Model Design Rationale*: The stacking architecture was chosen to reduce both bias and variance. Individual models capture different aspects of the data: linear models capture global trends, tree-based models capture nonlinear interactions, and neural networks capture high-order relationships.

The use of OOF predictions ensures unbiased training of the meta-learner, while hyperparameter tuning improves the performance of individual base models. The passthrough mechanism further enhances the model by incorporating both raw features and model predictions.

To further analyse the contribution of each base learner, an ablation study was conducted by removing individual models from the ensemble and evaluating performance changes. This provides insight into model importance and the effectiveness of model diversity within the stacking framework.

Overall, this design represents an enhanced stacking framework that goes beyond standard implementations by integrating optimisation, robustness, and interpretability considerations.

F. Evaluation Strategy

The performance of the proposed models was evaluated using a comprehensive framework that combines hold-out testing, cross-validation, hyperparameter tuning, and ablation analysis.

1) *Evaluation Metrics*: Two standard regression metrics were employed:

- **Root Mean Squared Error (RMSE)**: Measures the average magnitude of prediction errors and penalises larger errors more heavily. It provides an intuitive interpretation of prediction error in the same unit as house prices.
- **Coefficient of Determination (R^2)**: Measures the proportion of variance in the target variable explained by the model, indicating overall goodness-of-fit.

2) *Training and Validation Procedure*: The dataset was split into training and testing sets using an 80/20 ratio. All models were trained on the training set and evaluated on the test set to assess generalisation performance.

In addition, 5-fold cross-validation was applied during training to evaluate model robustness. For base models, cross-validation was used to compute average RMSE scores, providing a more stable estimate of performance.

For the stacking ensemble, a 5-fold Out-of-Fold (OOF) strategy was used to generate training predictions for the meta-learner. This ensures that the meta-model is trained on predictions obtained from unseen data, thereby preventing data leakage.

3) *Model Comparison*: To evaluate the effectiveness of the proposed approach, individual base models (Linear Regression, Random Forest, XGBoost, and MLP) were first trained and evaluated separately. Their performance was compared with the stacking ensemble to assess whether combining models leads to improved predictive accuracy.

Furthermore, an enhanced stacking variant with feature passthrough was evaluated, where both base model predictions and original input features were used as inputs to the meta-learner. This allows for a more flexible comparison between standard and extended stacking designs.

4) *Hyperparameter Optimisation Evaluation*: Hyperparameter tuning was performed using grid search with cross-validation, particularly for XGBoost. The best-performing parameter configuration was selected based on cross-validated RMSE, and the optimised model was then incorporated into the stacking framework.

5) *Ablation Study*: An ablation study was conducted to analyse the contribution of each base model within the stacking ensemble. This was achieved by systematically removing one base learner at a time and evaluating the resulting performance changes.

This analysis provides insight into model importance and demonstrates how different models contribute to the overall ensemble performance.

6) *Interpretability Analysis*: To further understand model behaviour, feature importance analysis was performed using Random Forest and SHAP (SHapley Additive exPlanations). These methods provide insights into how individual features influence predictions and help validate whether the model aligns with domain knowledge.

7) *Evaluation Rationale*: This evaluation framework ensures both reliability and fairness. The use of a hold-out test set provides an unbiased estimate of generalisation performance, while cross-validation improves robustness.

The inclusion of hyperparameter tuning, model comparison, and ablation analysis enables a comprehensive assessment of model effectiveness. In addition, interpretability analysis enhances transparency, making the results more meaningful in real-world contexts.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

V. DISCUSSION

VI. CONCLUSION AND FUTURE WORK

REFERENCES

- [1] N. Kok, P. Monkkonen, and J. M. Quigley, "Land use regulations and the value of land and housing: An intra-metropolitan analysis," *Journal of Urban Economics*, vol. 81, pp. 136–148, 2014. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0094119014000278>
- [2] S. Rosen, "Hedonic prices and implicit markets: Product differentiation in pure competition," *Journal of Political Economy*, vol. 82, no. 1, pp. 34–55, 1974. [Online]. Available: <http://www.jstor.org/stable/1830899>
- [3] V. Limsombunchao, "House price prediction: Hedonic price model vs. artificial neural network," *Agricultural Economics Research Unit Working Paper*, 2004.
- [4] G.-Z. Fan, S. E. Ong, and H. C. Koh, "Determinants of house price: A decision tree approach," *Urban Studies*, vol. 43, no. 12, pp. 2301–2315, 2006.
- [5] J. R. Rico-Juan and P. Taltavull de La Paz, "Machine learning with explainability or spatial hedonics tools? an analysis of the asking prices in the housing market in alicante, spain," *Expert Systems with Applications*, vol. 171, p. 114590, 2021. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0957417421000312>
- [6] A. Kumar, P. Jain, P. Pandey, S. Das, V. K. Gupta, and D. Kumar Gupta, "House price prediction system using ensemble learning approach," in *2024 Eighth International Conference on Parallel, Distributed and Grid Computing (PDGC)*, 2024, pp. 304–309.
- [7] Z. Li, S. Xie, Y. Zhang, and J. Hu, "A study on house price prediction based on stacking-sorted-weighted-ensemble model," *Journal of Internet Technology*, vol. 23, no. 5, pp. 1139–1146, 2022.
- [8] W. Kim and J. Hong, "Stacked ensemble model for the automatic valuation of residential properties in south korea: A case study on jeju island," *Land*, vol. 13, no. 9, 2024. [Online]. Available: <https://www.mdpi.com/2073-445X/13/9/1436>
- [9] E. Bjørgeve, A. Oust, A. J. Pollestad, C. Sandnes, and O. J. Sønstebo, "Comparing housing valuation techniques and stacked generalization: Exploiting explainable ai," *The Journal of Real Estate Finance and Economics*, vol. 72, pp. 640 – 665, 2025. [Online]. Available: <https://api.semanticscholar.org/CorpusID:281736517>
- [10] H. Selim, "Determinants of house prices in turkey: Hedonic regression versus artificial neural network," *Expert Systems with Applications*, vol. 36, no. 2, Part 2, pp. 2843–2852, 2009. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0957417408000596>
- [11] A. Hjort, J. Pensar, I. Scheel, and D. E. Sommervoll, "House price prediction with gradient boosted trees under different loss functions," *Journal of Property Research*, vol. 39, no. 4, pp. 338–364, 2022. [Online]. Available: <https://doi.org/10.1080/09599916.2022.2070525>
- [12] V. Cherkassky and Y. Ma, "Practical selection of svm parameters and noise estimation for svm regression," *Neural Networks*, vol. 17, no. 1, pp. 113–126, 2004. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0893608003001692>
- [13] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, no. 1, p. 5–32, Oct. 2001. [Online]. Available: <https://doi.org/10.1023/A:1010933404324>
- [14] J. H. Friedman, "Greedy function approximation: A gradient boosting machine," *The Annals of Statistics*, vol. 29, no. 5, pp. 1189 – 1232, 2001. [Online]. Available: <https://doi.org/10.1214/aos/1013203451>
- [15] D. H. Wolpert, "Stacked generalization," *Neural Networks*, vol. 5, no. 2, pp. 241–259, 1992. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0893608005800231>
- [16] S. Ganapathy and T. Li, "An explainable ai framework with ml model stacking for house price prediction," in *2024 IEEE MIT Undergraduate Research Technology Conference (URTC)*, 2024, pp. 1–5.